

Step 11: Solution Requirements

BookNest — Functional and Non-Functional Requirements

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through form (name, email, password) Role selection during registration (Customer / Seller)
FR-2	User Login & Authentication	Login using registered email and password Secure session handling using JWT tokens
FR-3	Book Browsing & Search	Browse books by category Search books by title or author View detailed book information (price, stock, description)
FR-4	Cart & Order Management (Customer)	Add or remove books from cart Place an order and complete checkout View order history and order status
FR-5	Book Listing Management (Seller)	Add a new book listing with details and stock Edit or update existing book details and stock Deactivate or remove a listing
FR-6	Order Fulfilment (Seller)	View incoming orders for listed books Update order or shipment status
FR-7	User & Seller Management (Administrator)	View all registered users and sellers Activate or deactivate a user/seller account Review or moderate book listings
FR-8	Platform Monitoring (Administrator)	Dashboard overview of users, sellers, books, and orders View summary of overall platform activity

Non-Functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface should be simple and intuitive for all three roles (Customer, Seller, Administrator), requiring minimal learning to perform common tasks.
NFR-2	Security	User passwords are stored using secure hashing, and access to routes and data is controlled through JWT-based authentication and role-based access control.
NFR-3	Reliability	The system should consistently maintain accurate book stock, order, and user data, with proper error handling to avoid data inconsistency.
NFR-4	Performance	Common actions such as browsing books, placing orders, and loading dashboards should respond within a few seconds under normal usage.

NFR No.	Non-Functional Requirement	Description
NFR-5	Availability	The application should be accessible whenever it is deployed/running, with minimal unplanned downtime affecting customers, sellers, or administrators.
NFR-6	Scalability	Built on a modular MERN-based architecture, the system should be able to accommodate growth in the number of users, sellers, and book listings without major redesign.